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Pre-Engineered Quarantine Centre Building

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ABSTRACT

In the face of global health challenges, the need for rapidly deployable and efficient quarantine facilities has become increasingly apparent. This paper presents a comprehensive analysis and design framework for a Pre-Engineered Quarantine Centre (PEQC) building. The proposed structure aims to provide a flexible and scalable solution that can be swiftly deployed in response to infectious disease outbreaks.

The analysis phase begins with a thorough examination of the unique requirements and constraints associated with quarantine facilities, considering factors such as infection control, ventilation systems, spatial arrangements, and adaptability to different site conditions. A risk assessment is conducted to identify potential hazards and vulnerabilities, informing the design process to prioritize safety and security.

The design phase integrates the principles of pre-engineering to develop a modular and prefabricated structure that optimizes construction time and cost while maintaining the necessary functionality for a quarantine facility. The building design incorporates state-of-the-art materials and technologies to ensure durability, energy efficiency, and compliance with relevant health and safety standards.

Key features of the proposed PEQC design include modular isolation units, dedicated zones for medical staff, efficient waste management systems, and advanced air filtration to prevent the spread of infectious agents. The layout prioritizes patient flow, minimizing cross-contamination risks and ensuring ease of medical surveillance.

Furthermore, the paper explores the integration of digital technologies for monitoring and managing the facility, including real-time data analytics, automated contact tracing, and telemedicine capabilities. These technological advancements enhance the overall efficiency and responsiveness of the quarantine center.

The proposed analysis and design framework provide a robust foundation for the development of Pre-Engineered Quarantine Centres, offering a versatile and adaptable solution to public health emergencies. The research contributes to the ongoing discourse on pandemic preparedness, emphasizing the importance of innovative and scalable infrastructure to safeguard public health on a global scale.

Keywords:- *Building, construction, Design, isolation,etc.*

INTRODUCTION

GENERAL

The World Wellbeing Association has expressed that as of sixth June 2021 there have been 172,630,637 affirmed instances of Coronavirus in the realm of which 37, 18, 683 passings have been accounted for to WHO. It is unsettling to bring up that even our state isn't falling behind in that Coronavirus details. Kerala has recorded a normal Test Energy Pace of 15.01 somewhat recently that implies for each

100 individuals tried 15 are viewed as sure. In this hazardously disturbing circumstance, our most importantly obligation is to guarantee the wellbeing of individuals. At the point when you take a gander at the world nations there are a few models like Hong Kong and South Korea who have enthusiastically battled the infection and controlled its spread generally by depending on separation or quarantine focuses show to the public authority. With this the need and significance of isolation focuses are deeply grounded.

After more than a year, nothing has changed; rather, everything has only gotten worse. With the remarkable expansion in the affirmed cases subsequently understanding the significance and need of independently lodging individuals to break the chain, in this undertaking we mean to design, examine, plan and gauge the expense of a pre-designed quarantine place building.

ADVANTAGES OF PEB

Following are some of the advantages Pre-Engineered Building Structures:

- The World Prosperity Affiliation has communicated that as of 6th June 2021 there have been 172,630,637 avowed cases of Covid in the domain of which 37, 18, 683 passings have been represented to WHO.
- It is disrupting to raise that even our state isn't falling behind in that Covid subtleties.
- Kerala has recorded an ordinary Test Energy Speed of 15.01 to some degree as of late that suggests for every 100 people attempted 15 are considered certain.
- Our primary responsibility is to ensure the well-being of individuals in this perilous and troubling circumstance.
- If you look at the countries around the world, you'll see a few examples like Hong Kong and South Korea that have actively fought the disease and slowed its spread generally by relying on quarantine or separation centers shown to the public authority.
- With this the need and meaning of seclusion centers are profoundly grounded.
- After over a year, nothing has changed; rather, everything has just deteriorated.
- With the momentous development in the certified cases hence understanding the importance and need of freely dwelling people to break the chain, in this endeavor we mean to configuration, look at, plan and measure the cost of a pre-planned quarantine place building.

OBJECTIVES

- To plan a quarantine centre with necessary requirements.
- To design the quarantine centre using Pre- Engineered Building technology.
- To design the connections manually.
- Cost estimation of materials.

METHODOLOGY

- Literature Survey.
- Data Collection.
- Software Study.
- Detailed Study of Design Guidelines.
- Drafting the plan of the Quarantine centre using AutoCad Software.
- Analysis and Design using Staad Software.
- Design of Connections manually.
- Foundation design using STAAD Foundation
- Cost estimation of materials.
- Final Report Preparation.

LITERATURE REVIEW

C. M. Meera made the observation in June 2013 that the Pre-Engineered Building (PEB) concept is a novel way to build single-story industrial buildings. This strategy is flexible not just because of its quality pre-planning and construction, yet in addition because of its light weight and practical development. The idea incorporates the strategy of giving the most ideal area as per the ideal necessity. Compared to the Conventional Steel Building (CSB) concept of buildings with roof trusses, this one offers numerous advantages. This paper is a near investigation of PEB idea and CSB idea. Pre-Designed Building idea have wide applications including stockrooms, manufacturing plants, workplaces, studios, corner stores, display areas, vehicle leaving sheds, airplane storages, metro stations, schools, sporting structures, indoor arena rooftops, outside arena shades, rail line stage covers, spans, halls and so on. PEB designs can likewise be planned as re-locatable designs.

Syed Firoz, Sarath Chandra Kumar et.al (2012) saw that, the pre-designed steel building framework development enjoys extraordinary benefits to the single story structures, reasonable and proficient option in contrast to traditional structures, the Framework addressing one focal model inside numerous disciplines. Pre-designed building makes and keeps up with continuously multi-faceted, information rich perspectives through an undertaking support is right now being carried out by Staad ace programming bundles for plan and designing. Picking steel to plan a Pre-designed steel structures building is to pick a material which offers minimal expense, strength, solidness, plan adaptability, versatility and recyclability.

Aijaz Ahmad Zende, Prof. A. V. Kulkarni, et.al (Jan. - Feb. 2013) saw that despite the fact that PEB structures gives clear range, it weighs lesser than that of Customary Structures. Steel is the material that exemplifies the requirements of sustainable development because it is infinitely recyclable. Pre-designed building are the best answer for longer range structures with practically no inside in the middle between. With the approach of computerization, the plan prospects turned out to be practically boundless. Saving of material on low pressure region of the essential outlining individuals makes Pre-designed structures more practical than Ordinary steel structures particularly for low ascent structures crossing up to 90.0 meters with eave levels up to 30.0 meters. To put it plainly, Pre-Designed Building Development gives the end clients a significantly more practical and improved answer for long range structures where huge segment free regions are required.

Jatin D. Thakar, Prof. P.G.Patel (Sep 2013) saw that, Pre-designed building are steel building wherein the outlining individuals and different parts are completely manufactured in the processing plant in the wake of planning and brought to the site for gathering, basically by nut-fasteners, in this manner coming about into a steel construction of great and accuracy. In ordinary steel development, we have site welding included, which isn't true in PEB utilizing nut-bolt component. These designs utilize hot moved tightened areas for essential outlining and cold moved segments (by and large Z and C areas) for auxiliary outlining according to the inner pressure prerequisites, subsequently decreasing wastage of steel and oneself load of the construction and thus lighter establishments.

G. Sai Kiran, A. Kailasa Rao, R. Pradeep Kumar (Aug 2014) saw that, lately, the presentation of Pre Designed Building (PEB) idea in the plan of designs has assisted in upgrading with planning. The adoption of PEB as an alternative to the conventional steel building (CSB) design concept brought about a number of benefits, including cost savings and simplified fabrication. In this review, a modern construction (Product House) is broke down and planned by the Indian guidelines, IS 800-1984, IS 800-2007 and furthermore by alluding MBMA-96 and AISC-89.

S.D. Charkha and Latesh S (June 2014) saw that, utilizing of PEB rather than CSB might be decreasing the steel amount. Decrease in the steel amount certainly lessening the dead burden. Decrease in the dead burden diminishing the size of Establishment. Utilizing of PEB increment the tasteful perspective on structure.

SOFTWARES USED

AutoCAD

AutoCAD is a PC Helped Plan (computer aided design) program utilized for 2-D and three dimensional plan and drafting. AutoCAD is strong programming authorized via Autodesk. AutoCAD is utilized for drawing various designs, subtleties, plans, heights and various segments. It is an exceptionally valuable programming for common, mechanical and electrical specialists and is likewise utilized in different enterprises by modelers, project directors, engineers, visual creators, city organizers and numerous different experts.

Advantages of using AutoCAD are:

- One of the fundamental benefits of the computer aided design configuration administrations is the capacity to control and alter the plan easily.
- One more benefit is the capacity to store and save the documents in a concentrated data set with the end goal of future adjustment and record keeping.
- The utilization of computer aided design configuration administration has likewise made present day and complex structures with much better plan and usefulness.
- Saving the expense and having the computerized documents prepared whenever.
- AutoCAD gives exceptional drafting devices that are utilized in drawings of designing parts, plans and framework.
- Limits human mistakes and assists the clients with rejuvenating their creative mind with exactness.

STAAD PRO V8i

STAAD Genius is an underlying examination and plan programming which is generally used to investigate and configuration structures for spans, towers, structures, transportation, modern and utility designs. It is a high level 3D underlying investigation programming authorized by Bentley. STAAD Genius is a primary programming, acknowledged by underlying specialists that settles an ordinary static investigation of the breeze, seismic examination with different mixes of burden to adjust different codes, for example, IS 456:2000, IS 1893:2002, IS 875:1987 and so on.

In STAAD, we can make a model, dole out the property of the materials, relegate the heaps that follow up on the design and the product examinations them and creates a report with support subtleties. The product has now its most recent form, STAAD Ace V8i with new better elements. Any engineering structure can now be analyzed and designed using the STAAD Pro V8i.

Advantages of using STAAD Pro V8i are:

- STAAD Master show precise outcomes in estimation of shear power and bowing second.
- It includes no manual computation subsequently saves time and increments effectiveness.

Because of its built-in parametric library, we are able to modify it to fit any design we require.

- This product helps engineers in working on the design, area and aspects.
- STAAD Genius offers quicker techniques for planning the construction.
- Cement and steel configuration have been incorporated into STAAD Star to assist us with upgrading our plan with full control boundaries, for example, redirection, support for substantial segments, pillars, chunks and shear walls and subsequently, the pressure of utilizing one programming to do displaying, one more programming for steel plan and another to configuration cement footers, section and establishment is cleared out.

STAAD FOUNDATION

STAAD. Establishment is an independent programming with in excess of 50 different plan modules and 5 unique global codes. Its expense saving downstream application that likewise empowers architects to examine and plan the basic starting point for the construction they made in STAAD.Pro. This software is used to design the foundation.

MS EXCEL

The nitty gritty gauge for the structure according to the plan is finished with the assistance of MS Succeed. The expense for the base works like earthwork uncovering, establishment, earth filling and so on, are displayed in the part Assessment.

GUIDELINES FOR QUARANTINE CENTRE

The rules for setting up of isolation offices during the Coronavirus flare-up is given by public focus to infectious prevention under the service of wellbeing and family government assistance, Administration of India. The suggested span of isolation for Coronavirus in light of accessible data is upto 14 days from the hour of openness. During the current outbreak, quarantine is being used to reduce transmission by

- Separating contacts of COVID-19 patients from community.
- Monitoring contacts for development of sign and symptoms of COVID-19.
- Segregation of COVID-19 suspects, as early as possible from among other quarantined persons.

Requirements for Quarantine facility in a community-based facility is as under

1. Location:

- Preferably placed in the outskirt of the urban/ city area (can be a hostel/unused health facilities/buildings, etc.) away from the people’s reach, crowded and populated area.
- Well protected and secured (preferably by security personnel/ army).
- Preferably should have better approachability to a tertiary hospital facility having critical care and isolation facility.

2. Access considerations:

- Parking space including Ambulances etc.
- Ease of access for delivery of food/medical/other supplies
- Differently-abled Friendly facilities (preferably)

3. Ventilation capacity: Well ventilated preferably natural

4. Basic infrastructure/functional requirements:

- Rooms/Dormitory separated from one another may be preferable with in-house capacity of 5-10 beds/room.
- Each bed to be separated 1-2 meters (minimum 1 metre) apart from all sides.
- Lighting, well-ventilation, heating, electricity, ceiling fan.
- Potable water to be available.
- Functional telephone system for providing communications.

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- Support services- fooding, snacks and recreation areas including television.
- Laundry services.
- Sanitation services/Cleaning and Housekeeping.
- Properly covered bins as per BMW may be placed

5. Space requirements for the facility:

- Administrative offices- Main control room/clerical room
- Logistics areas/Pharmaceutical rooms.

ARCHITECTURAL PLAN

The architecture plan of the quarantine centre building is shown in Fig1 .

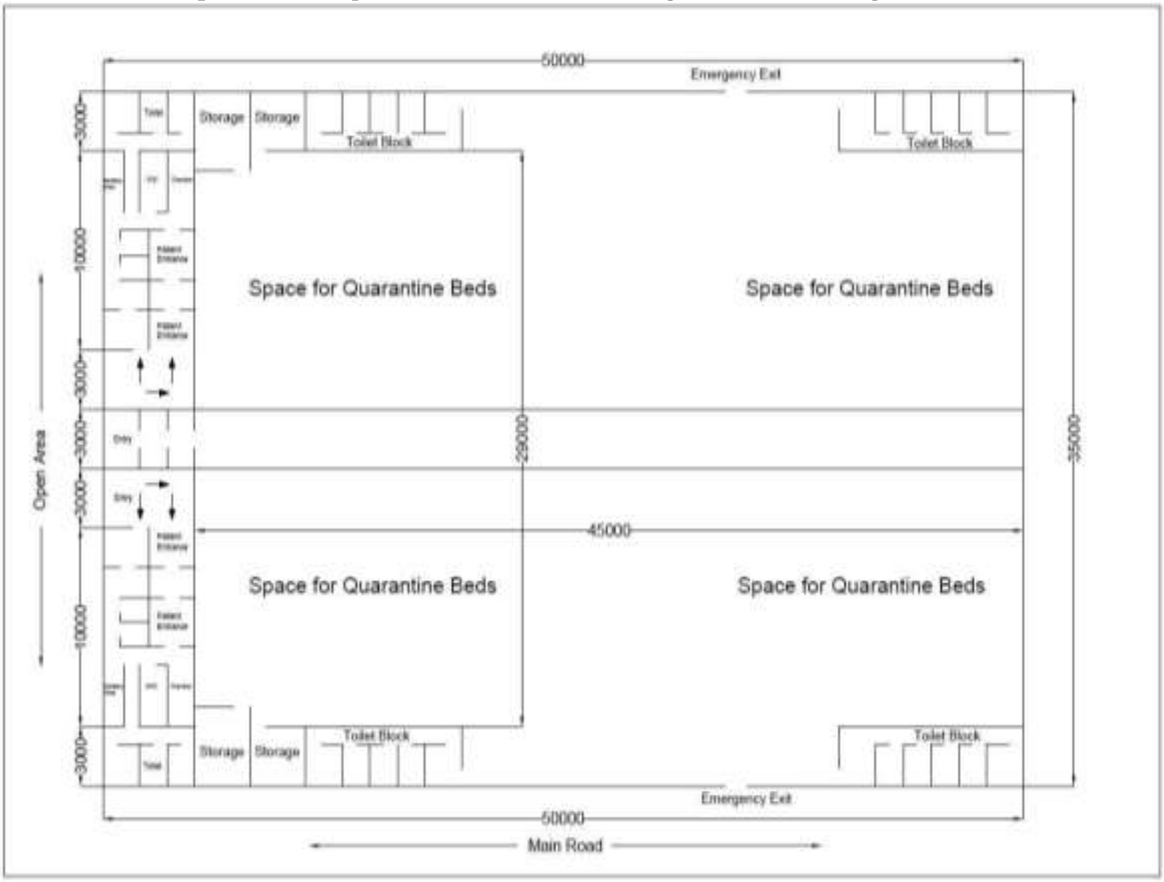

***Fig.1**:-Plan of Quarantine Centre Building*

Building details:

Building Length (L) –50 m
Building Width (W) –35 m
Area = 19000 sq ft
Bay Spacing -10 m
Clear height -7 m
Roof Slope -6.5 degree
Purlin spacing -1.8m c/c
Roofing material -Galvanised steel corrugated sheets
Wall coverings -GI/aluminium sheet 0.8 mm thick
Partition walls -12.5 mm thick Gypsum plaster board Number of beds: 145

LOAD CALCULATIONS

Dead Load
Dead loads are considered as per Table 2 of IS 875 (Part-1)-1987.The dead load is primarily due to the self-weight of structure and the roofing materials.
Weight of roofing materials = 0.2 kN/m²
Live loads are considered as per Table 2 of IS 875 (Part-2)-1987
Live load on the sloping roof = 0.75 kN/m²
Bay spacing = 10 m
Live load on rafter = 0.75 x 10 = 7.5 kN/m
Live load
Live loads are considered as per Table 2 of IS 875 (Part-2)-1987
Live load on the sloping roof = 0.75 kN/m²
Bay spacing = 10 m
Live load on rafter = 0.75 x 10 = 7.5 kN/m

Wind Load
Wind loads are determined according to IS 875 Section III (2015). The essential breeze speed (Vb) is 39m/s for Kerala and the structure is viewed as open landscape with all around dispersed deterrents having level under 10m with most extreme aspect more than 50m and likewise factors K1, K2, K3 have been determined according to IS 875 Section III (2015).
Probability factor, K1= 1.0
Terrain, height and size factor, K2 =0.91
Topography factor, K3 =1
Design wind speed, Vz = Vb (K1 x K2 x K3) Vz = 35.4 m/s
Design pressure, P= 0.6 Vz^2 = 0.272 kN/m²
Design Wind Force, $F_d = C_f A_e P_d$
F_d = 0.35 KN/m²

Seismic Load
Since the building is located in Kerala, as per IS 1893-2016 (Part-1) it is included in Zone III and the load parameters considered are as follows:
Z = 0.16, Zone factor given in Table 3.
I = 3, Importance factor given in Table 8.
R = 3, Response reduction factor given in Table 9.
SS = 3, Rock and soil site factor.
DM = 0.05, Damping ratio.

Load Combinations
Load combinations were taken according to IS 456-2000 and IS 1893 (Part 1)-2002. The following are the combinations taken for the analysis: -
1. DL+LL
2. DL+WL
3. 1.2 (DL + LL + WL)
4. 1.5 (DL + LL)
5. 0.9DL + 1.5WL
6. DL + 0.8LL + 0.8 WL
7. DL + 0.8LL + 0.8EQ(+X)
8. DL + 0.8LL + 0.8EQ(-X)
9. DL + 0.8LL + 0.8EQ(+Z)

10. DL + 0.8LL + 0.8EQ(-Z)
11. 1.5(DL + EQ(+X))
12. 1.5(DL + EQ(-X))
13. 1.5(DL + EQ(+Z))
14. 1.5(DL + EQ(-Z))
15. DL + EQ(+X)
16. DL + EQ(-X)
17. DL + EQ(+Z)
18. DL + EQ(-Z)
19. 0.9DL + 1.5EQ(+X)
20. 0.9DL + 1.5EQ(-X)
21. 0.9DL + 1.5EQ(+Z)
22. 0.9DL + 1.5EQ(-Z)
23. 1.2DL +0.3LL +1.2EQ(+X)
24. 1.2DL +0.3LL +1.2EQ(-X)
25. 1.2DL +0.3LL +1.2EQ(+Z)
26. 1.2DL +0.3LL +1.2EQ(-Z)

ANALYSIS IN STAAD Pro

IS 875 Section III is used to calculate wind loads. The fundamental breeze speed (Vb) is 39m/s for Kerala and the construction is seen as open scene with overall around scattered impediments having level under 10m with most outrageous perspective more than 50m and in like manner factors K1, K2, K3 not entirely settled by IS 875 Area III (2015). 3D view of the model is shown in Figure 2 and rendered one in Figure 3.

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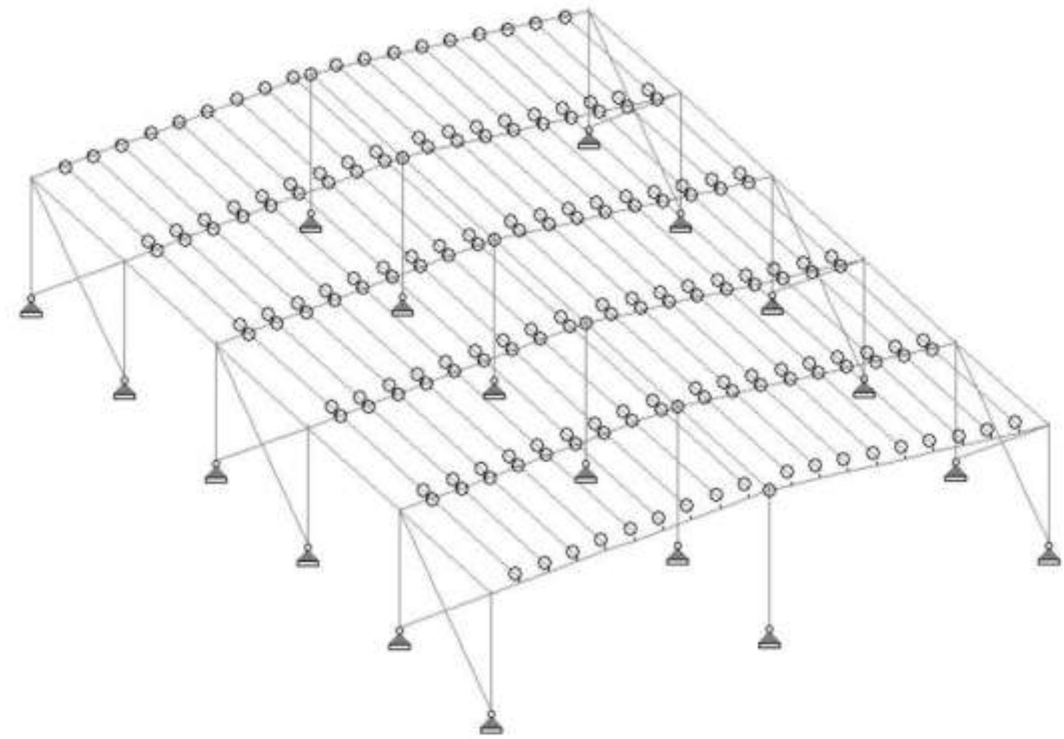

Fig.2:*-3D View of Model*

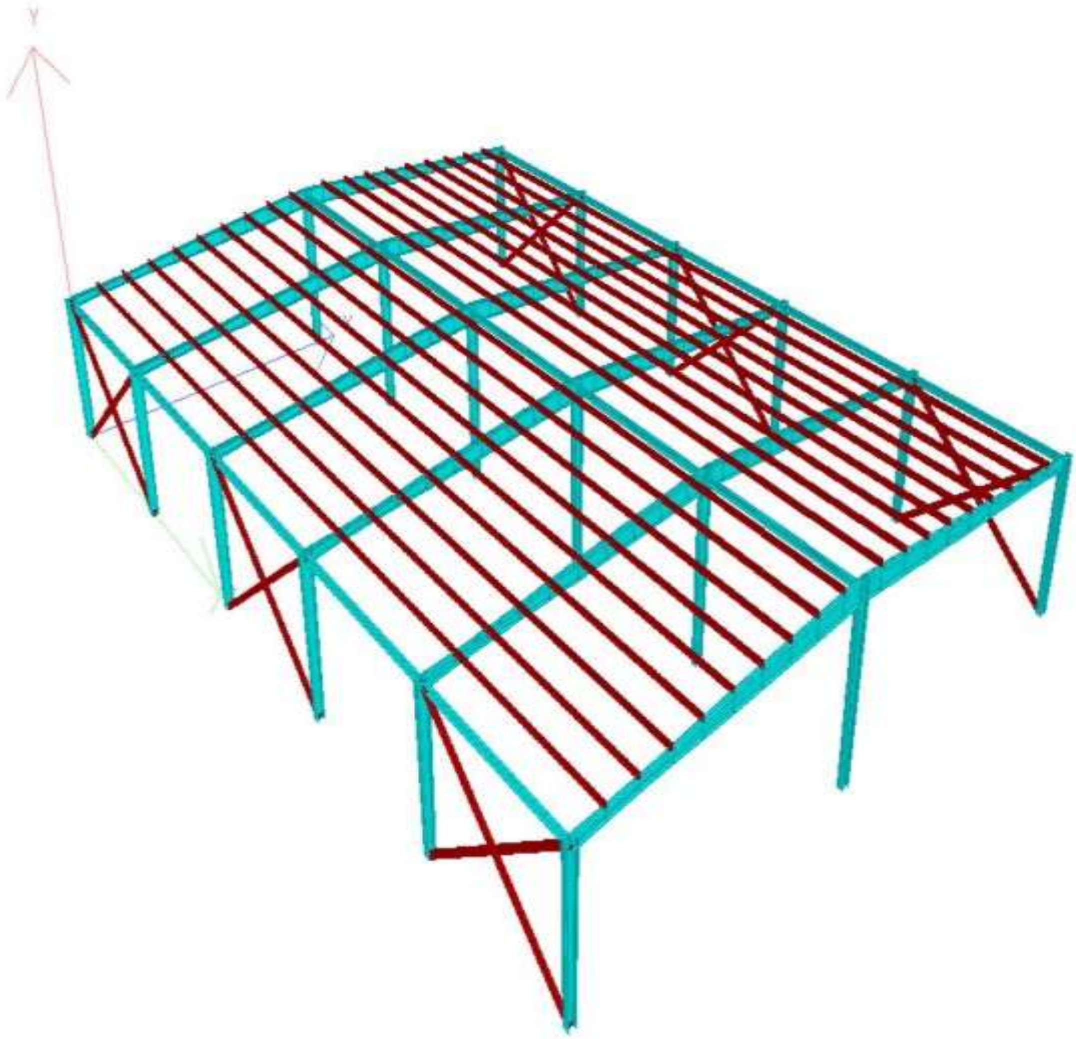

Fig.3:-Rendered View of the Model in STAAD Pro

The design utilize hot moved tightened segments for essential outlining and cold moved areas for auxiliary outlining according to the inward pressure prerequisites, in this manner diminishing wastage of steel and oneself load of the construction and consequently lighter establishments. Essential framings utilized are second opposing edges with stuck base and cold shaped C segments are utilized as auxiliary framings. The primary members' cross sections, which are tapered in accordance with specifications, are provided below.

CROSS SECTION DETAILS OF TAPERED COLUMN:

Depth of section at start node, F1 = 400 mm
Thickness of web, F2 = 16mm
Depth of section at end node, F3 = 590mm
Width of top flange, F4 = 350 mm
Thickness of top flange, F5 = 16 mm
Width of bottom flange, F6 = 350 mm
Thickness of bottom flange, F7 = 16 mm

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CROSS SECTION DETAILS OF TAPERED BEAM:

BEAM 1
Depth of section at start node, F1 = 450 mm
Thickness of web, F2 = 10mm
Depth of section at end node, F3 = 350mm
Width of top flange, F4 = 300 mm
Thickness of top flange, F5 = 10 mm
Width of bottom flange, F6 = 300 mm
Thickness of bottom flange, F7 = 10 mm

BEAM 2
Depth of section at start node, F1 = 350 mm
Thickness of web, F2 = 10mm
Depth of section at end node, F3 = 650mm
Width of top flange, F4 = 300 mm
Thickness of top flange, F5 = 10 mm
Width of bottom flange, F6 = 300 mm
Thickness of bottom flange, F7 = 10 mm

BEAM 3
Depth of section at start node, F1 = 650 mm
Thickness of web, F2 = 10mm
Depth of section at end node, F3 = 850mm
Width of top flange, F4 = 300 mm
Thickness of top flange, F5 = 10 mm
Width of bottom flange, F6 = 300 mm
Thickness of bottom flange, F7 = 10 mm

Figure 4 shows the Cross Sectional Detail and the Figure 5 shows the Bending moment diagram
Shear force diagram and Axial force diagramme shown in Figure 6 & 7 respectively

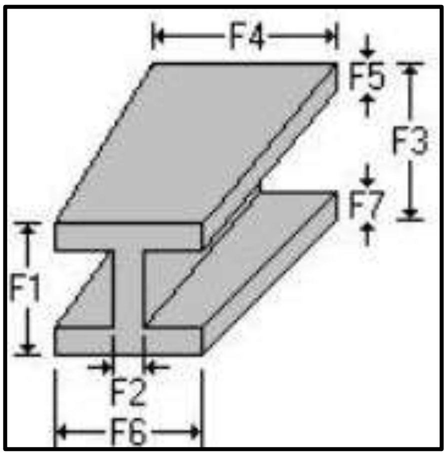

Fig.4:-*Cross Sectional Detail*

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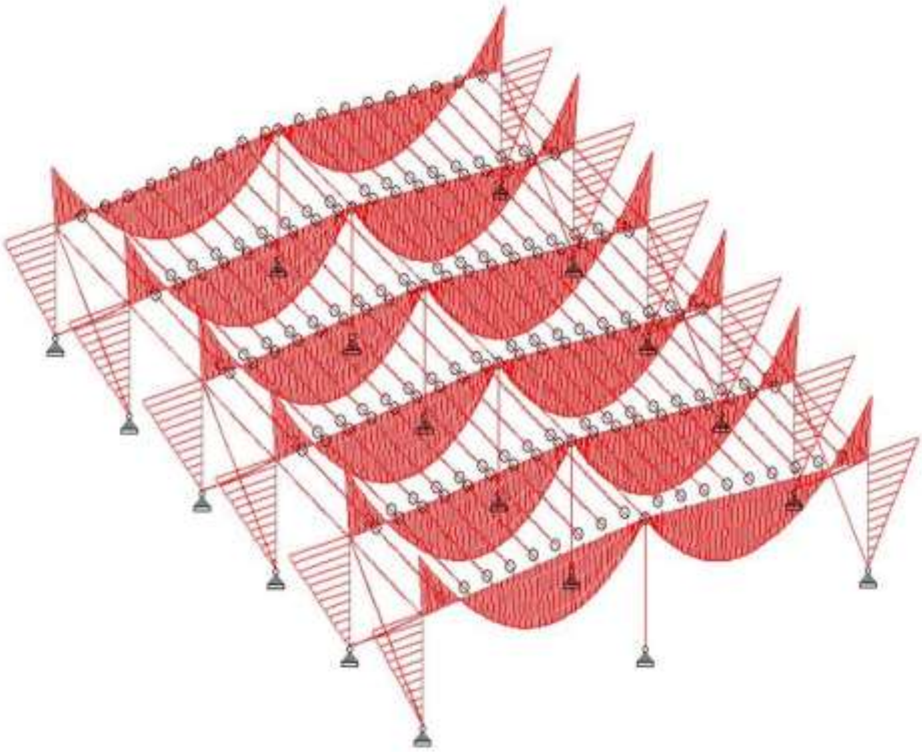

Fig.5:- *Bending moment diagram*

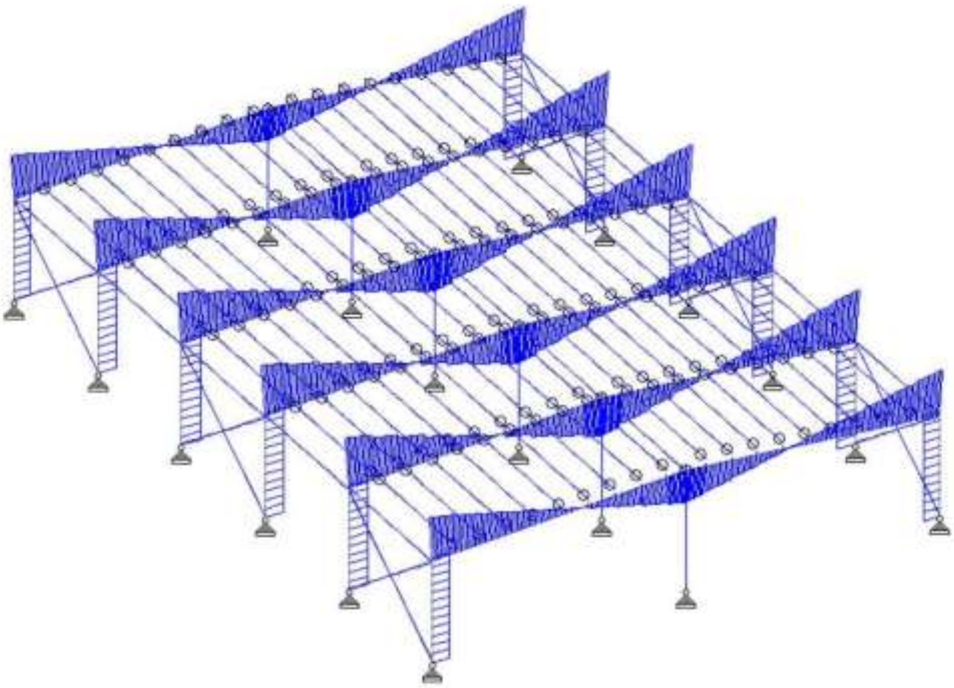

Fig.6:-*Shear force diagram*

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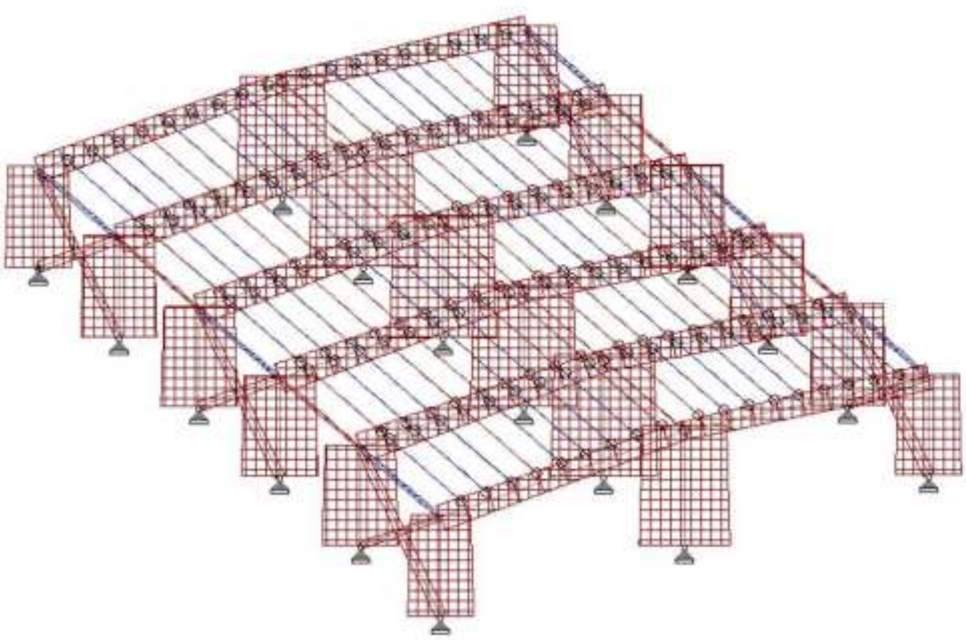

***Fig.7:-** Axial force diagram*

- The maximum value of shear shear force is 179.12 kN and is found in load case 1.5(DL+LL).
- The maximum value of bending moment is 481.38 kNm and is found in load case 1.5(DL+LL).
- The maximum value of axial force is 280.49 kN and is found in load case 1.5(DL+LL).

DESIGN OF CONNECTIONS

DESIGN OF COLUMN BASE PLATE AND ANCHOR BOLT CONNECTION

Depth of section, D = 400 mm
Depth of web = 368 mm
Thickness of web, tw = 16 mm
Thickness of flange, tf =16 mm
Flange width, bf =350 mm
Axial load = 280 kN
Shear force = 70 kN
Grade of concrete = M25 MPa
Bearing strength of concrete = 0.45* fck = 0.45*25 = 11.25 N/mm²

Anchor bolt details

Diameter of anchor bolt, d = 24 mm
Total number of anchor bolt = 8
Gross area of bolt, Asb = 452.16 mm²
Net area of bolt = 352.7 mm²
Grade of bolt = 8.8
Ultimate tensile strength of bolt, Fub = 830 MPa
Yield stress of bolt, Fyb = 660 MPa

Shear check:

Factored shear force in bolt, Vsb ≤ Shear design strength of bolt, Vdsb
Vsb = Shear force/n
Assume 8 number of bolts
Vsb = 70/8 = 8.75 kN
Vdsb = Vnsb/Ymb
Vnsb = $\frac{fub}{\sqrt{3}}(n_n$ Anb+n_s Asb) = 169.014 kN
Vdsb = 135.21 kN
Vsb < vdsb. Hence safe

Tensile check:
Factored tensile force in bolt ≤ Tensile strength of bolt
Factored tensile forec in bolt, 'Tb' = 280/4 = 70 kN
Tdb = Tnb/Ymb
Tnb = 0.9*fub*An < fyb*Asb*Ymb/Ymo
Ymb= 1.25, Ymo = 1.1
0.9*fub*An = 263.467 kN
fyb*Asb*Ymb/Ymo = 339.12 kN
Considering the minimum value, Tnb = 263.467 kN
Tdb = Tnb/Ymb = 263.467/1.25 = 210.77 kN
Since Tb ≤ Tdb, safe

Combined shear and tension check:
$(Vsb/Vdb)^2 + (Tb/Tdb)^2 \leq \mathbf{1}$
$(15/135.21)^2 + (85/210.77)^2$ = 0.457 < 1, Hence ok

Calculation of anchor bolt length
For M25 concrete, bond strength in limit state method for plain bars = τbd = 1.4 N/mm²
(As per IS 456:2000 - clause 26.2.1.1)
Anchor length required = Tb/π*d* τbd
= 663.48 mm
Therefore, provide anchor bolt of length 700 mm with dia 24 mm.

Base plate details
Ultimate tensile strength of plate, Fu = 410 MPa
Yield stress of plate, Fv = 240 MPa
Size of base plate provided = 500 mm × 450 mm
Area of base plate provided = 225000 mm²
Factored load = 1.5*280
= 420 kN
Required area = 420*1000/bearing strength
= 37333.33 mm² < Area provided, Hence ok
Therefore, Length of base plate = 500 mm
Width of base plate = 450 mm

a) Under downward force
Bearing pressure under base plate, W = P/A
= 1.5*280*1000/500*450
=1.867N/mm² < fck

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Thickness of base plate, t = $\sqrt{(2.5 * w * (a^2 - 0.3b^2)\ Ymo/fv)}$

(as per IS 800 2007 - clause 7.4.3.1)

W =1.867N/mm²
a = (500 – (368+ 16 + 16))/2 = 50 mm
b = (450 – 350)/2 = 50 mm
Ymo = 1.25
fv = 240 MPa
Therefore, t = 6.52 mm, take 8 mm

b) Under uplift force

Z = (b*t²)/6
Therefore, Plate thickness, t = (6*Z)/b
b = width of base plate/2
= 450/2 = 225 mm
Edge distance, e = 1.7*Diameter of bolt hole = 1.7*26
= 50mm
Yield stress of plate, Fv = 240 N/mm²
Z = M/Fv
M = Tb*e = 3500000N-mm
Design stress in plate, Fd = 1.2*Fv/1.1
Fd = 261.818
Hence, Z = M/Fd = 13368.055 mm³
Therefore thickness, t = $\sqrt{(6 * Z)/b}$
= 18.88 mm, take 20 mm
Take maximum of the two values, thickness of base plate = 20 mm
Provide 8, 24 mm dia anchor bolt of length 700 mm
Provide Base plate of thickness = 20 mm.
Detailing of baseplate and anchor bolt connection is shown as Fig 8

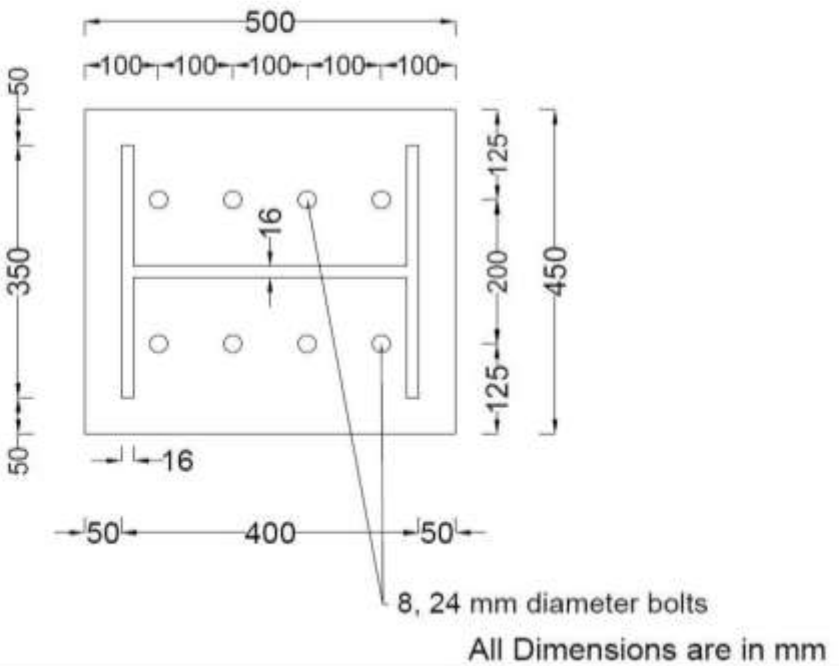

Fig.8:-Detailing of baseplate and anchor bolt connection.

BEAM - BEAM CONNECTION DESIGN
Nominal diameter of anchor bolt, db = 24 mm
Gross area of bolt, Asb = 452.16 mm²
Net area of the bolt,Anb = 0.78*Gross Area = 352.68 mm²
Assume 8 number of bolts
Depth of connected beam d = 650 mm
Width of beam flange, bf = 300 mm
Thickness of beam web, tw = 10 mm
Shear force, V = 20 kN
Bending moment, M = 410 kN-m
Tensile axial force, T = 60 kN
Grade of bolt =8.8
Ultimate tensile strength of bolt, fub =830 MPa
Yield stress of bolt, fyb =660 MPa

BOLT DESIGN
Shear check:
Factored shear force in bolt, Vsb ≤ Shear design strength of bolt, Vdsb
Vsb = Shear force/n
Assume 8 number of bolts
Vsb = 2.5 kN
Vdsb = Vnsb/Ymb
Vnsb = $\frac{fub}{\sqrt{3}}(n_n$ Anb+n_s Asb) = 162.89 kN
Vdsb = 135.20 kN
Vsb ≤ Vdsb, Hence safe.
Tensile check:
Factored tensile force in bolt ≤ Tensile strength of bolt
Flange force = (M/d) + (T/2)
Where M = bending moment
D = depth of beam
T = Tensile axial force
Flange force = 660.77 KN

Tension per bolt, Tb = Flange force/4
Tb = 165.192 kN
Tdb = Tnb/Ymb
Tnb = 0.9*fub*An < fyb*Asb*Ymb/Ymo
Ymb = 1.25, Ymo = 1.1
0.9*fub*An = 263.54 kN
fyb*Asb*Ymb/Ymo = 339.12 kN
Considering the minimum value, Tnb = 263.45 kN
Tdb = Tnb/Ymb
= 210.76kN
Since Tb ≤ Tdb, safe.

Combined shear and tension check:
$(Vsb/Vdb)^2 + (Tb/Tdb)^2$ **≤ 1**

$(2.5/135.20)^2 + (165.19/210.76)^2$ = 0.614< 1, Hence OK
Hence provide 8, 24 mm dia bolts.

COLUMN – BEAM CONNECTION DESIGN
FRAME 1:
Nominal diameter of anchor bolt, db = 30mm
Gross area of bolt, Asb =706.mm²
Net area of the bolt, Anb =551.07mm²
Assume 8 number of bolts
Depth of connected beam d = 450 mm
Width of beam flange, bf = 300 mm
Thickness of beam web, tw =10 mm
Shear force, V = 155 kN
Bending moment, M = 410 kN m
Tensile axial force, T = 160 kN
Grade of bolt =8.8
Ultimate tensile strength of bolt, fub = 830 MPa
Yield stress of bolt, fvb = 660 MPa

BOLT DESIGN
Shear check:
Factored shear force in bolt, Vsb ≤ Shear design strength of bolt, Vdsb
Vsb = Shear force/n
Assume 8 number of bolts
Vsb = =155/8 = 19.375 KN
Vdsb = Vnsb/Ymb
Vnsb = $\frac{fub}{\sqrt{3}}(n_n$ Anb+n_s Asb) = 264.073 kN
Vdsb = 264.073/1.25 = 211.2585 kN
Vsb <=Vdsb, Hence safe.
Tensile check:
Factored tensile force in bolt ≤ Tensile strength of bolt
Flange force = (M/d) + (T/2)
Where M = bending moment
D = depth of beam
T = Tensile axial force
Flange force = (410*100/0.45) + (160/2)
= 991.11 kN -m
Tension per bolt, Tb = Flange force/4
Tb = 991.11/4 = 247.77 kN
Tdb = Tnb/Ymb
Tnb = 0.9*fub*An < fyb*Asb*Ymb/Ymo
Ymb = 1.25, Ymo = 1.1
0.9*fub*An = 0.9*830*551.07 =411.64kN
fyb*Asb*Ymb/Ymo = 660*314*1.25/1.1 = 529.875 kN
Considering the minimum value, Tnb = 411.64 kN
Tdb = Tnb/Ymb
=411.64/1.25 = 329.32 kN
Since Tb <= Tdb, safe.

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Combined shear and tension check:
$(Vsb/Vdb)^2 + (Tb/Tdb)^2 \leq \mathbf{1}$
= $(19.375/211.25)^2 + (247.77/329.32)^2$
= 0.574 < 1, Hence OK
Provide 8, 30 mm dia bolts.

FRAME 2:
Nominal diameter of anchor bolt, db = 30 mm
Gross area of bolt, Asb = 706.5 mm²
Net area of the bolt, Anb = 551.07 mm²
Assume 8 number of bolts
Depth of connected beam d = 450 mm
Width of beam flange, bf = 300 mm
Thickness of beam web, tw = 10 mm
Shear force, V = 180 kN
Bending moment, M = 480kN
Tensile axial force, T = 185 kN
Grade of bolt = 8.8
Ultimate tensile strength of bolt,fub = 830 MPa
Yield stress of bolt, fvb = 660 MPa

BOLT DESIGN
Shear check:
Factored shear force in bolt, Vsb ≤ Shear design strength of bolt, Vdsb
Vsb = Shear force/n
Assume 8 number of bolts
Vsb = 180/8 = 22.5 kN
Vdsb = Vnsb/Ymb
Vnsb = $\frac{fub}{\sqrt{3}}(n_n$ Anb+n_s Asb) = 264.07kN
Vdsb = =264.07/1.25 = 211.25 kN
Vsb < vdsb, hence safe.
Tensile check:
Factored tensile force in bolt ≤ Tensile strength of bolt
Flange force = (M/d) + (T/2) Where M = bending moment=400kNm
D = depth of beam = 450mm
T = Tensile axial force =170kN
Flange force = 1159.167kNm
Tension per bolt, Tb = Flange force/n
Tb = 1159.167/4 = 289.79 kN
Tdb = Tnb/Ymb
Tnb = 0.9*fub*An < fyb*Asb*Ymb/Ymo
Ymb = 1.25, Ymo = 1.1
0.9*fub*An = 411.649 kN
fyb*Asb*Ymb/Ymo = 529.875kN
Considering the minimum value, Tnb = 411.649 kN
Tdb = Tnb/Ymb
= 411.649/1.25 =329.32 kN
Since Tb <= Tdb, safe.

Combined shear and tension check:
$(Vsb/Vdb)^2 + (Tb/Tdb)^2$ **≤ 1= 0.785 < 1,** Hence ok.
Provide 8, 30 mm dia bolts.

FRAME 3:
Nominal diameter of anchor bolt, db = 30mm
Gross area of bolt, Asb = 706.5 mm²
Net area of the bolt, Anb =551.07 mm²
Assume 4 number of bolts
Depth of connected beam d = 450mm
Width of beam flange, bf = 300mm
Thickness of beam web, tw = 10mm
Shear force, V =180 kN
Bending moment, M = 480 kN-m
Tensile axial force, T = 186 kN
Grade of bolt = 8.8
Ultimate tensile strength of bolt,fub = 830 MPa
Yield stress of bolt, fvb = 660 MPa

BOLT DESIGN
Shear check:
Factored shear force in bolt, Vsb ≤ Shear design strength of bolt, Vdsb
Vsb = Shear force/n
Assume 8 number of bolts
Vsb = 180/8 = 22.5 kN
Vdsb = Vnsb/Ymb
Vnsb = $\frac{fub}{\sqrt{3}}(n_n$ Anb+n_s Asb) = 264.07 kN
Vdsb = 264.07/1.25 = 211.258 kN
Vsb < vdsb, Hence safe.
Tensile check:
Factored tensile force in bolt ≤ Tensile strength of bolt
Flange force = (M/d) + (T/2)
Where M = bending moment = 390 kN-m
D = depth of beam = 450 mm
T = Tensile axial force = 155 kN
Flange force = 1159.667 kN -m
Tension per bolt, Tb = Flange force/n
Tb = 1159.667/4 = 289.917 kN
Tdb = Tnb/Ymb
Tnb = 0.9*fub*An < fyb*Asb*Ymb/Ymo
Ymb = 1.25, Ymo = 1.1
0.9*fub*An = 411.649 kN
fyb*Asb*Ymb/Ymo = 529.875 kN
Considering the minimum value, Tnb = 411.649 kN
Tdb = Tnb/Ymb
=411.649/1.25 = 329.32 kN
Since Tb <= Tdb, Hence safe.
Combined shear and tension check:

$(Vsb/Vdb)^2$+ $(Tb/Tdb)^2$ **≤ 1 = 0.786 < 1**, hence safe.
Provide 8, 30 mm dia bolts.

8.4 BEAM - COLUMN CONNECTION DESIGN (Shear connection)
Nominal diameter of anchor bolt, db = 16 mm
Gross area of bolt, Asb = 200.96 mm^2
Net area of the bolt, Anb = 0.78*Gross Area = 156.748 mm^2
Assume 4 number of bolts
Depth of connected beam d = 850 mm
Width of beam flange, bf = 300 mm
Thickness of beam web, tw = 10 mm
Shear force, V = 130 kN
Grade of bolt = 8.8
Ultimate tensile strength of bolt,fub = 800 MPa
Yield stress of bolt, fyb = 640 MPa

BOLT DESIGN
Shear check:
Factored shear force in bolt, Vsb ≤ Shear design strength of bolt, Vdsb
Assume 4 number of bolts
Vsb = Shear force/4
Vsb = 32.5 kN
Vdsb = Vnsb/Ymb
Vnsb = $\frac{fub}{\sqrt{3}}(n_n$ Anb+n_s Asb) = 75.114 kN
Vdsb = 60.09 kN
Vsb ≤Vdsb, hence safe.
Therefore, provide 4, 16 mm dia bolts.
Beam Column Connection is shown in Fig 9

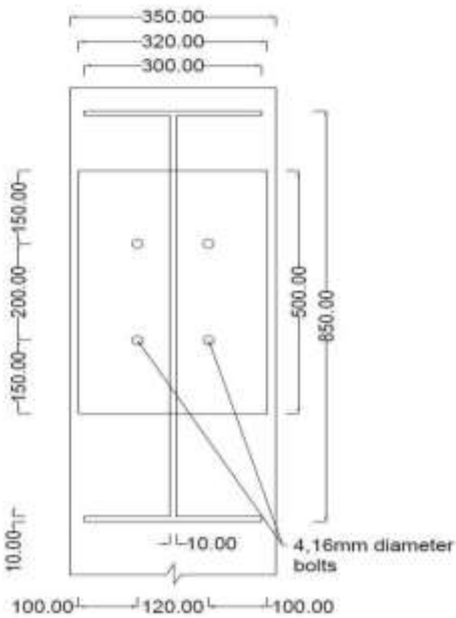

***Fig.9:**-Beam Column Connection*

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DESIGN OF FOUNDATION

Plan of establishment is finished utilizing the product STAAD Establishment. The balance calculation, balance fortifications and their spacings are given in light of the plan results got from STAAD establishment. The sloped isolated footing reduces the amount of concrete required, thereby lowering the cost of the footing. A sum of eighteen footings are given. Table 1 shows the elements of different establishments. Table 2 shows the support subtleties of the footings.

CONCRETE AND REBAR PROPERTIES

Unit weight of concrete = 25 kN/m³
Strength of concrete = 25 N/mm²
Yield strength of steel = 415 N/mm²
Minimum bar size = 10 mm
Maximum bar size = 32 mm
Minimum bar spacing = 50 mm
Maximum bar spacing = 500 mm
Footing clear cover = 50 mm
Footing thickness = 400 mm
Slope end thickness = 200 mm
Column length = 600 mm
Column width = 550 m
Plan and elevation of footing are shown as Fig 10

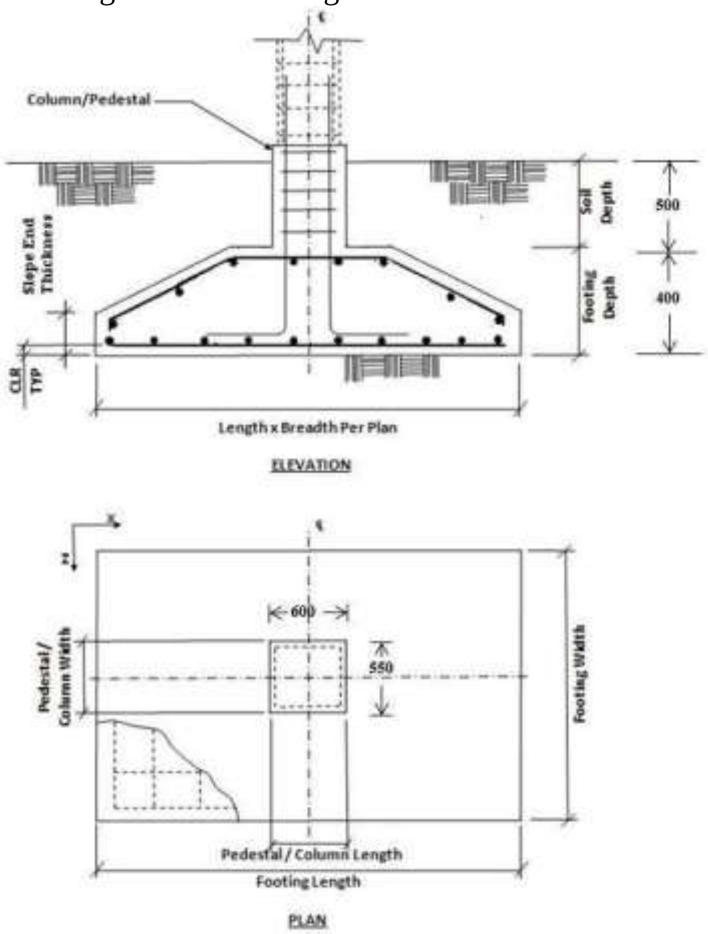

Fig.10:- Plan and elevation of footing

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Foundation Geometry and Footing Reinforcement is detailed in Table 9.1 & Table 9.2 respectively

Table 1:-*Foundation Geometry*

FOOTING NO.	GROUP ID	FOUNDATION GEOMETRY			
-	-	**Length**	**Width**	**Thickness**	**Slope End Thickness**
1	1	1.650 m	1.600 m	0.400 m	0.200 m
2	2	1.650 m	1.600 m	0.400 m	0.200 m
5	3	1.950 m	1.900 m	0.400 m	0.200 m
25	4	1.550 m	1.500 m	0.400 m	0.200 m
26	5	1.550 m	1.500 m	0.400 m	0.200 m
29	6	1.550 m	1.500 m	0.400 m	0.200 m
49	7	1.850 m	1.800 m	0.400 m	0.200 m
50	8	1.850 m	1.800 m	0.400 m	0.200 m
53	9	1.550 m	1.500 m	0.400 m	0.200 m
73	10	1.650 m	1.600 m	0.400 m	0.200 m
74	11	1.650 m	1.600 m	0.400 m	0.200 m
77	12	1.550 m	1.500 m	0.400 m	0.200 m
97	13	1.750 m	1.700 m	0.400 m	0.200 m
98	14	1.750 m	1.700 m	0.400 m	0.200 m
101	15	1.550 m	1.500 m	0.400 m	0.200 m
121	16	1.600 m	1.550 m	0.400 m	0.200 m
122	17	1.600 m	1.550 m	0.400 m	0.200 m
125	18	1.950 m	1.900 m	0.400 m	0.200 m

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Table 2:-*Footing Reinforcement*

FOOTING NO.	**FOOTING REINFORCEMENT**			
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)
1	Ø10 @ 165 mm c/c	Ø10 @ 150 mm c/c	Ø10 @ 165 mm c/c	Ø10 @ 150 mm c/c
2	Ø10 @ 165 mm c/c	Ø10 @ 150 mm c/c	Ø10 @ 165 mm c/c	Ø10 @ 150 mm c/c
5	Ø10 @ 160 mm c/c	Ø10 @ 165 mm c/c	Ø10 @ 160 mm c/c	Ø10 @ 165 mm c/c
25	Ø10 @ 150 mm c/c	Ø10 @ 160 mm c/c	Ø10 @ 150 mm c/c	Ø10 @ 160 mm c/c
26	Ø10 @ 150 mm c/c	Ø10 @ 160 mm c/c	Ø10 @ 150 mm c/c	Ø10 @ 160 mm c/c
29	Ø10 @ 150 mm c/c	Ø10 @ 160 mm c/c	Ø10 @ 150 mm c/c	Ø10 @ 160 mm c/c
49	Ø10 @ 165 mm c/c	Ø10 @ 155 mm c/c	Ø10 @ 165 mm c/c	Ø10 @ 155 mm c/c
50	Ø10 @ 165 mm c/c	Ø10 @ 155 mm c/c	Ø10 @ 165 mm c/c	Ø10 @ 155 mm c/c
53	Ø10 @ 150 mm c/c	Ø10 @ 160 mm c/c	Ø10 @ 150 mm c/c	Ø10 @ 160 mm c/c
73	Ø10 @ 165 mm c/c	Ø10 @ 150 mm c/c	Ø10 @ 165 mm c/c	Ø10 @ 150 mm c/c
74	Ø10 @ 165 mm c/c	Ø10 @ 150 mm c/c	Ø10 @ 165 mm c/c	Ø10 @ 150 mm c/c
77	Ø10 @ 150 mm c/c	Ø10 @ 160 mm c/c	Ø10 @ 150 mm c/c	Ø10 @ 160 mm c/c
97	Ø10 @ 155 mm c/c	Ø10 @ 160 mm c/c	Ø10 @ 155 mm c/c	Ø10 @ 160 mm c/c
98	Ø10 @ 155 mm c/c	Ø10 @ 160 mm c/c	Ø10 @ 155 mm c/c	Ø10 @ 160 mm c/c
101	Ø10 @ 150 mm c/c	Ø10 @ 160 mm c/c	Ø10 @ 150 mm c/c	Ø10 @ 160 mm c/c
121	Ø10 @ 160 mm c/c	Ø10 @ 165 mm c/c	Ø10 @ 160 mm c/c	Ø10 @ 165 mm c/c
122	Ø10 @ 160 mm c/c	Ø10 @ 165 mm c/c	Ø10 @ 160 mm c/c	Ø10 @ 165 mm c/c
125	Ø10 @ 160 mm c/c	Ø10 @ 165 mm c/c	Ø10 @ 160 mm c/c	Ø10 @ 165 mm c/c

COST ESTIMATION OF MATERIAL

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Cost estimation of materials is done using MS Excel. The rates are taken with reference to DSR 2018 by CPWD.The Cost Estimation of Materials is shown in Table 10.1 follows

Table 10.*1: Cost Estimation of Materials*

ABSTRACT						
SL NO	**DESCRIPTION**	**QUANTITY**	**UNIT**	**RATE**	**AMOUNT**	**REMARK**
1	Earth work in excavation by mechanical means (Hydraulic excavator)/manual means over areas (exceeding 30 cm in depth, 1.5 m in width as well as 10 sqm on plan) including getting out and disposal of excavated earth lead upto 50 m and lift upto 1.5 m, as directed by Engineer-in-charge. All kinds of soil.	44.541	cum	181.9	8099.781	DSR 2018
2	Providing and laying in position specified grade of reinforced cement concrete, excluding the cost of centering, shuttering, finishing and reinforcement - All work up to plinth level: 1:1.5:3 (1 cement: 1.5 coarse sand (zone-III): 3 graded stone aggregate 20 mm nominal size).	18.301	cum	7718	141251.7	
3	Supplying and filling in plinth with sand under floors, including watering, ramming, consolidating and dressing complete.	26.24	cum	1953	51248.03	
4	Steel reinforcement for R.C.C. work including straightening, cutting, bending, placing in position and binding all complete upto plinth level. Hot rolled deformed bars.	640	kg	83.5	53440	
5	Structural steel work riveted,	83000	kg	101.8	8445250	

	bolted or welded in built up sections, trusses and framed work, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer all complete.					
6	Providing and fixing bolts including nuts and washers complete.	10	kg	128.5	1284.5	
7	Galvanised steel corrugated sheets.	120	quintal	5600	672000	
8	Ceramic Glazed Tiles 1st quality 300 x 300 in all shades designs except White, Ivory, Grey, Fume Red Brown etc.	1750	sqm	260	455000	
9	GI/Aluminium Sheet (0.8 mm thick).	10000	kg	55	550000	
10	12.5 mm thick Gypsum plaster board.	1500	sqm	170	255000	
	TOTAL AMOUNT				**10632574**	

CONCLUSION

The preparation, examining, planning and assessment of Pre-designed quarantine place building is finished. This task assisted with acquiring openness about Pre-designed building innovation. The examination was finished utilizing the product bundle STAAD.ProV8i and the attracting subtleties AutoCAD 2018.The plan of associations were done physically as per important IS particular. The plan of establishment was finished utilizing STAAD establishment and assessment was finished utilizing MS succeed. Pre-Engineered Steel Buildings are superior to conventional buildings in terms of mass housing during emergencies due to their quicker construction. In this way a pre-designed quarantine community working with wellbeing, usefulness and solidness standards is demonstrated, dissected and planned.

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